Learning Location, Not Looking: A Placement-Memorization Audit and Ten-Episode Repair at the 30M Scale

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Abstract

Fixed-placement benchmarks let imitation policies learn *location* instead of *looking*. We document this on LIBERO-Object, whose target start poses are exactly pinned (measured) in 6 of 10 tasks and within $\sim \pm 1$ cm in the rest. First, an audit finds placement memorization at three layers: a calibrated expert whose offset constant encodes pose (sign-flipping under a software rebuild), an iteration-coupled selection loop (0.700 dev vs. 0.300 held-out), and a learned grasp head whose regression rationally prefers proprioception to pixels. Second, a repair—ten teacher episodes under ± 4 cm placement teleports plus jitter on a nuisance input—restores grounding (attribution: vision 1.1–1.8 cm, proprioception 0.1 cm), verified by a substitution probe paired with a behavioural randomization protocol: the released head scores 0.400 dev, 0.700 held-out, 0.400 randomized, with both non-dev protocols confirmed at n=50 (35/50 $[0.56, 0.81]$; 26/50 $[0.39, 0.65]$). The protocol's twin control cells scope that verification from both sides: on the deployment stack the memorized head scores 1/10 under the repaired head's own shift draws (where it scores 4/10), while on a rebuilt stack whose detector shift disables the repaired head's visual goals (0/10), the shell's search absorbs the same shifts for the proprioception-shortcut head (6/10). The behavioural certificate is therefore joint in (head, stack). Third, the policy class that made this reachable is goal-structured decoding driving an engineered pick-and-place shell; free per-tick regression scored 0/56. All results are simulation, one task and one object in every claimed cell, n=10 per cell and final at that n. A dated addendum carries the multi-object campaign: four objects attempted, a second crossed and twice confirmed on never-scored seeds, two measured, mechanism-located at role binding, and *not* crossed—with two pre-registered predictions recorded and falsified, and four instruments built to quantify the boundary all abandoned at their own calibration gates.

1 Introduction

A policy that scores 0.700 on a manipulation benchmark has, on the face of it, learned to see, decide, and act. This paper is about how much of such a number can instead be *location*: constants—in a benchmark's init files, in an expert's calibration, in a learned regressor's shortcut—that encode where the object always is, so that vision only ever gates an approach a memorized value finishes. On LIBERO-Object [10] we measure the preconditions exactly: the target object's start pose is bit-identical across all 50 shipped init states in 6 of 10 tasks, and varies by only 0.25–0.58 cm (std) in the other four. A benchmark with $\lesssim$1 cm of target variance admits position-memorizers, and we found them in every layer of our own system.

The vehicle is MicroVLA, a deliberately small language-conditioned vision-language-action stack: $\sim$30M parameters deployed, with a frozen open-vocabulary detector serving as the only vision *and* text encoder. The size is the paper's setting, not its claim. Its methodological value is that a stack this small has no capacity to paper over systems defects, so every seam had to be instrumented—and the instruments are what this paper contributes.

Contributions.

1. **A placement-memorization audit and its verification method.** A forensically worked instance of shortcut learning [6, 7] on a fixed-placement benchmark, shown at three layers, with the corrected measured account of what LIBERO-Object pins (per-task table and shipped measurement script); and

the probe-plus-behavioural-randomization protocol, motivated by the on-manifold limit of substitution probes and by a counterexample to probe-only certification. The protocol's twin control cells sharpen its scope: behavioural randomization separates memorized from repaired heads under identical draws on the deployment stack (1/10 vs. 4/10), and fails to separate them on a stack whose rebuild disables visual goals (6/10 vs. 0/10).

2. **A controlled policy-class result.** With perception, trunk, and corpus regime fixed, free per-tick regression scores 0/56 (upper 95% bound 6.4%) while goal-space supervision driving an engineered pick-and-place shell scores 4/10 dev, 7/10 held-out, 4/10 randomized—claimed strictly as a bundle, with the shell's share measured on both stacks.

3. **Training–serving-skew instrumentation.** 29 worked defect instances at producer/consumer seams, split into disagreements catchable by parity testing and agreements-on-a-wrong-convention catchable only by provenance [3, 14].

2 Platform

Table 1: Deployed parameter ledger. "Deployed" counts every parameter loaded at inference, frozen or trained.

component	params	status
YOLO-World-S detector (vision *and* the only text encoder)	13.0 M	frozen
Tiny Recursive Model world model	9.97 M	frozen at deployment
fusion + drift + relational + planner trunk heads	7.0 M	trained ($\leq$9 M cap)
goal heads (grasp 0.17 M + place 0.07 M) + learned gates	0.24 M	trained
deployed total	**≈30 M**	

No separate language model exists anywhere in the stack: the command is parsed into source/target phrases, and CLIP embeddings for (command, source, target) are harvested once per task from the detector's own text tower. **The world model is causally inert in every nonzero number of this paper**—a persistence-TRM ablation reproduces the held-out cell exactly (0.700).

3 Results

Table 2: Protocol × head. All cells $k/10$ with Wilson 95% intervals; task 0, wrist camera. Audit-stack cells are never pooled with deployment-stack cells (§ stack finding).

protocol	memorized head	**flagship (released)**	sibling A	sibling B
dev (states 0–9)	7/10	4/10	7/10	7/10
held-out (states 10–19)	3/10	**7/10**	7/10	6/10
randomized ±4 cm (same draws)	**1/10**	**4/10**	5/10	3/10
randomized ±4 cm (audit stack)	6/10	0/10	—	—

Confirmed at power: held-out **35/50** $= 0.700$ $[0.56, 0.81]$; randomized **26/50** $= 0.520$ $[0.39, 0.65]$. Across a detector-stack rebuild the repaired head reproduces in all three protocols while the memorized head's dev cell collapses 7/10 → 2/10 to its held-out floor: memorization is stack-coupled, grounding is what survived.

4 The multi-object addendum, and a falsified prediction

A second object (butter) crosses: 10/10 under an early-latch configuration, 6/10 with soup at 4/10 under a single anchor-band configuration with no per-object constants, and two independent pre-registered fresh-seed confirmations (5/10 + 3/10; 5/10 + 4/10 on a three-object head). Two further objects (cream cheese, chocolate pudding) do not cross, and the mechanism is located rather than asserted: goals are accurate where measurable (cream 1.58 cm on-manifold, vs. 1.34 and 1.06 for the objects that work) while *role binding* fails identically on both — each teacher never leaves its visual-servo phase, at source-uv std ∼0.30, settling tens of centimetres from the target. Four objects were attempted; two crossed.

Table 3: Role-binding sub-study: four binders, three objects, n=10 each, one head and one configuration. Offline identity accuracy is leave-episode-out 3-way classification on corpus crops.

binder	offline identity acc.	cream	butter	soup
none	—	0/10	4/10	5/10
crop-CLIP rerank	not measured	0/10	**7/10**	5/10
mean prototype (centered)	0.613	0/10	2/10	**8/10**
1-NN bank (centered)	0.902	0/10	4/10	6/10

Offline binder accuracy dissociates from deployed success. The weaker binder owns the best soup cell and the worst butter cell; the stronger returns butter to baseline; the binder with no offline number wins butter. A matched-episode A/B probe explains it: with the 0.902 binder active the uv stream fed to the grasp head *destabilises* (std 0.258 → 0.335 laterally, 0.047 → 0.202 vertically)—a binder accurate on corpus crops thrashes on live crops. This is the paper's own on-manifold limit recurring on two further instruments: a probe that evaluates on teacher trajectories cannot certify off-manifold behaviour, and a binder *built* from teacher trajectories cannot bind off-manifold crops.

Two predictions were pre-registered here and both are recorded as falsified: that 0.82 per-tick identity accuracy would yield 0.914 median-of-3 latch correctness and hence cream > 0 under bank binding (the cell returned 0/10), and that chocolate pudding would cross because it rarely won a misbind (its teacher never reached a grasp in five calibration iterations). Quantifying the deployed side of this boundary was attempted four times and abandoned four times, each at a pre-registered calibration gate: a projected-origin ground truth (rejected — soup, which succeeds 35/50, scored 0.111), its sign-corrected successor (rejected — the in-frame filter deletes the close-up target by construction), and three text-tower discrimination probes (rejected — each failed to detect the working object's own phrase). Deployed binding accuracy and text-tower discriminability are therefore reported as **unmeasured**, not estimated. The error is attributable—the prediction assumed offline per-tick accuracy equals deployed per-tick accuracy, which the probe refutes and which we do not measure. The prediction, its falsification, and this attribution are retained rather than removed.

5 Related work

Size is comparable only under a stated convention, and the two conventions in common use disagree about which system is smallest, because published counts routinely exclude a frozen language encoder that inference nonetheless loads. We therefore report both (Table 4). Under *trained parameters*, Octo-small's 27 M transformer [13] is the smallest published alternative and MicroVLA's 17.2 M is 1.6× below it; under *deployed* = every parameter loaded at inference, RT-1's 35 M [4] is the smallest published alternative and MicroVLA's 30.2 M is below it *even after granting RT-1 a free language encoder*—RT-1's 35 M excludes the Universal Sentence Encoder that produces its instruction embedding, for which we cite no count rather than estimate one. MicroVLA is thus the smallest of the surveyed systems under both conventions simultaneously, which is the strongest form of the claim the evidence supports.

The claim is bounded and checkable: it ranges over language-conditioned VLA policies with published parameter counts that we surveyed, it is a statement about parameter counts alone, and it is refuted by exhibiting any surveyed-class system below 17.2 M trained or 30.2 M deployed. We cannot rule out unpublished or unsurveyed smaller systems. The mechanism behind the gap is architectural rather than incidental: MicroVLA carries no separate language model at all, because the detector's own text tower supplies every text embedding—so the frozen encoder that the other rows pay for twice (once in memory, once in the convention mismatch) does not exist here. Of MicroVLA's 17.2 M trained parameters only 7.24 M are causally load-bearing, since the 9.97 M world model is inert in every nonzero number of this paper. No number here is comparable to community-protocol LIBERO scores (600 steps, n=10, wrist-only, scripted-expert

Table 4: Size under both conventions. "Trained" counts parameters fit by the authors; "deployed" counts every parameter loaded at inference, frozen or not. † excludes a frozen language encoder the published count omits. Sizes only: capability, task breadth, and protocol are not comparable.

system	trained	deployed	language encoder
RT-2 [5]	55 B	55 B	internal (PaLI-X)
OpenVLA [9]	7 B	7 B	internal (Llama-2)
SmolVLA [16]	450 M	450 M	internal (SmolVLM2)
Octo-small [13]	27 M	138 M	T5-base, 111 M frozen
RT-1 [4]	35 M	$\geq$35 M†	USE, frozen, uncounted
MicroVLA	**17.2 M**	**30.2 M**	*none*—detector's own tower

corpora). Predicting *where* rather than *how* is the Transporter [18] and CLIPort [15] lineage, continued by keypoint interfaces over engineered primitives [8, 11] and, with both levels learned, $\pi_{0.5}$ [2]. Our audit is a worked manipulation instance of causal confusion and copycat shortcuts [1, 6, 17]; what we add is the audited counterexample to probe-only certification, the resulting probe-plus-randomization pairing, and the control cells that scope the pairing itself. Bootstrapping from scripted experts is established [12]; our delta is that the scripted teacher is itself audited, caught encoding placement and detector version, and repaired.

6 Claims, non-claims, limitations

Claimed: the pinning measurement in its scoped form; the released head's cells above; free per-tick regression at 0/56; the de-memorization mechanism at the head level (attribution shift, varied-label validation, and a paired jitter ablation, 0/10 $\rightarrow$ 7/10 dev, exact McNemar/sign $p \approx 0.016$), scoped by the audit-stack control; 29 worked skew defects; and the bounded size result of Table 4 (smallest of the surveyed systems under both the trained and the deployed convention simultaneously). **Not claimed:** any task-success contribution from the world model (causally inert in every nonzero number); end-to-end learned motor control (the shell is engineered); head-level grounding from the randomized column alone, stack-free; object-level generalization beyond the addendum's two crossed objects (four attempted, two blocked at role binding); any deployed binding-accuracy figure; physical deployment; any size claim beyond the bounded one of Table 4 (surveyed systems, published counts, parameters only)—in particular no claim that smaller systems do not exist, and no claim that small size caused any success rate reported here. **Limitations:** n=10 per cell with n=50 confirmations on two protocols; ten held-out states reused across five disclosed selection looks; machine constants inherited from a dev-tuned era; substitution probes are on-manifold instruments; $\pm$4 cm randomization sits inside the shell's $\pm$6 cm probe envelope and separates heads only where visual goals are live; the teacher's calibration consumed privileged diagnostic telemetry offline; single benchmark, single simulator.

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